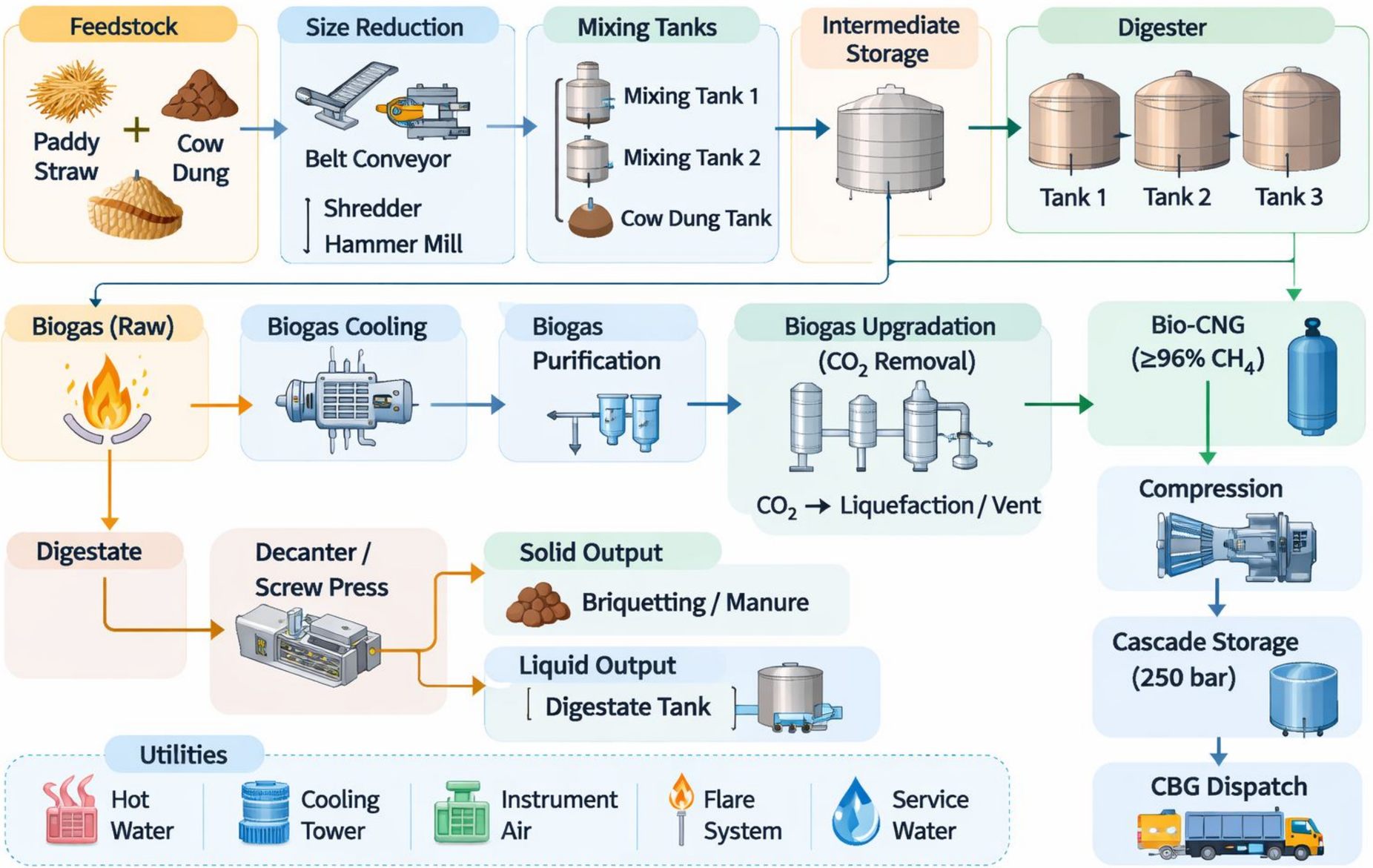
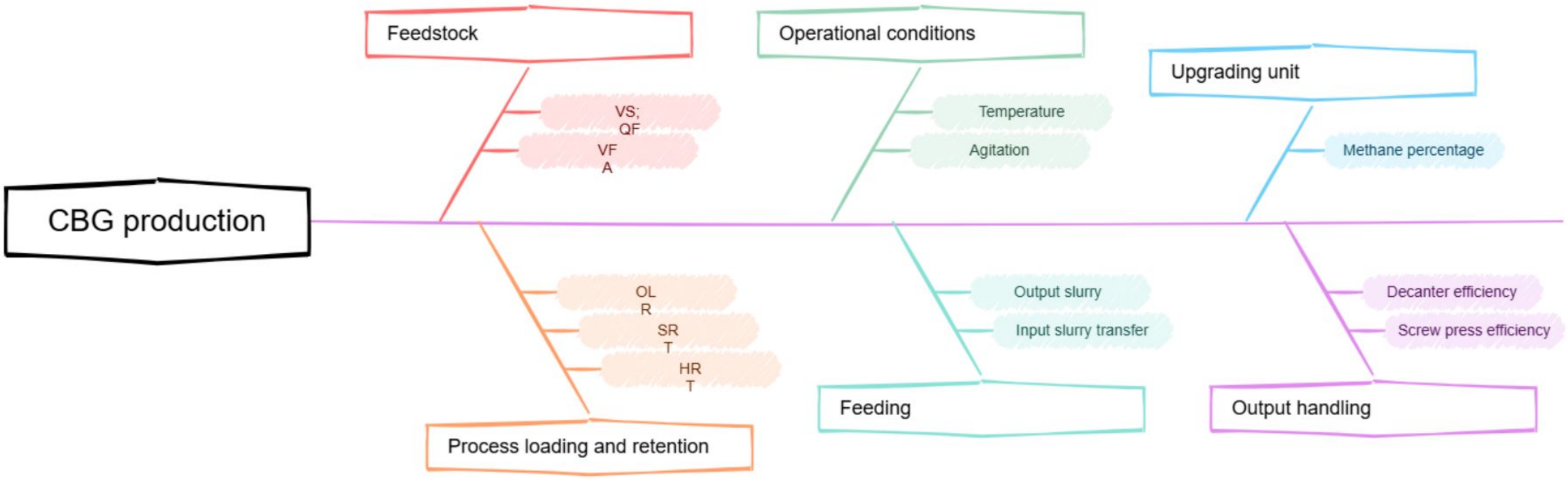
CBG Prediction Model

Rahneesh Mukkil - 13 Mar 2026

Paddy straw to Energy (CBG)



Features matters the most !



80:20 Rule

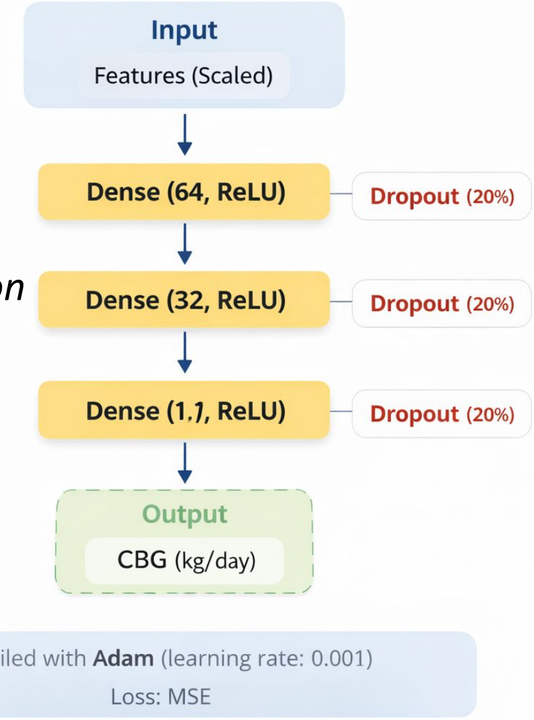
Features missed or unable to collect due to technical difficulties (Its very costly to analyse all these)

- Agitator current, power failures, equipment breakdowns, feedstock quality issues, Abrupt change in the client demand due to logistic pressure, heat loss in the boiler and insulation, Many other parameters like vibration, poor operation discipline, efficiency of each instruments and gauges.

Model - 1 : ANN

Relu introduces non linearity

Dense connect all the nodes

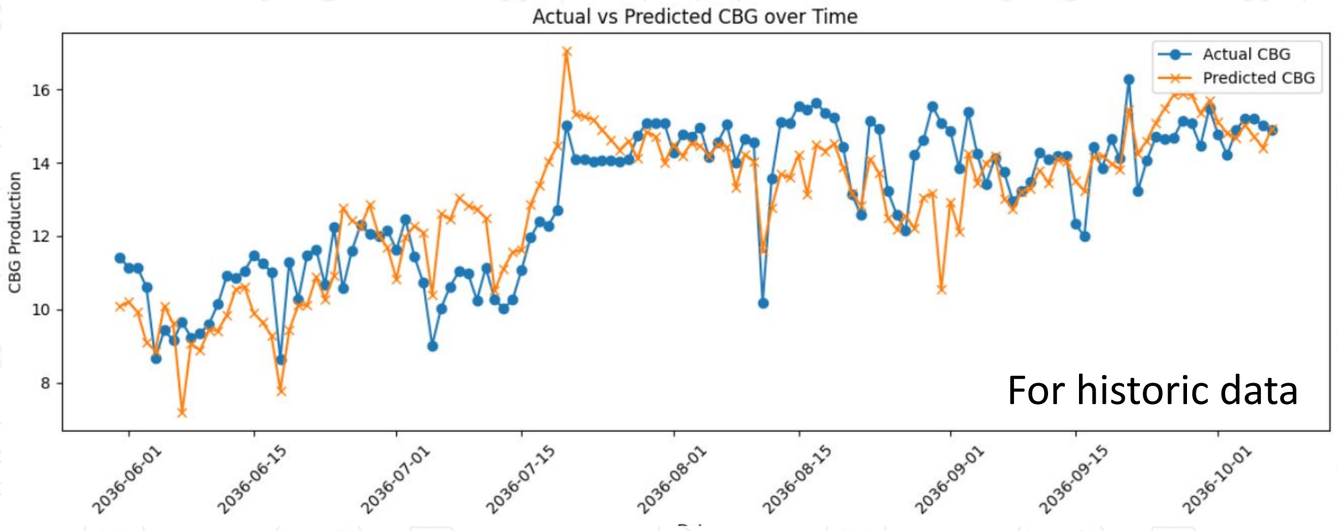


Derived Features (How ANN worked for a time series data)

➔ Lagging features - Feedstock feeding, slurry feeding, Temperatures (1 day, 3 day, 7 day)

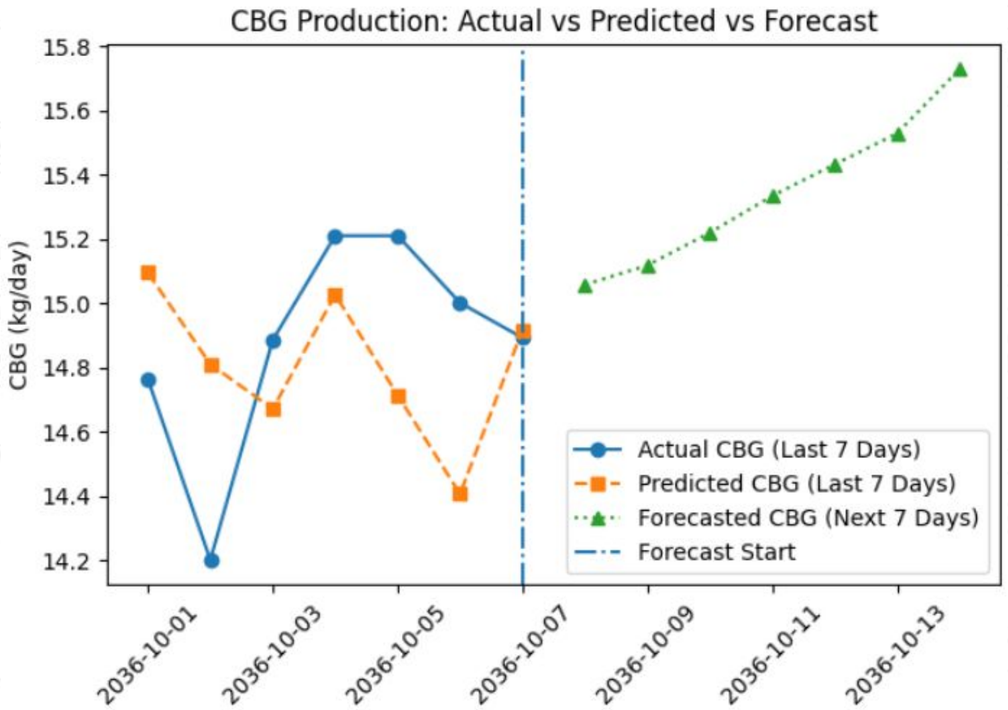
➔ MA features - Feedstock feeding, slurry feeding, Temperatures (1 day, 3 day, 7 day)

Result



Prediction ->

- Constant feeding
- No abnormal operation of equipments
- No power failure



Model - 2 : PROPHET

Prophet

- Daily Seasonality: Disabled
- Weekly Seasonality: Disabled
- Yearly Seasonality: Disabled

Interval Width: 0.90

Regressors:

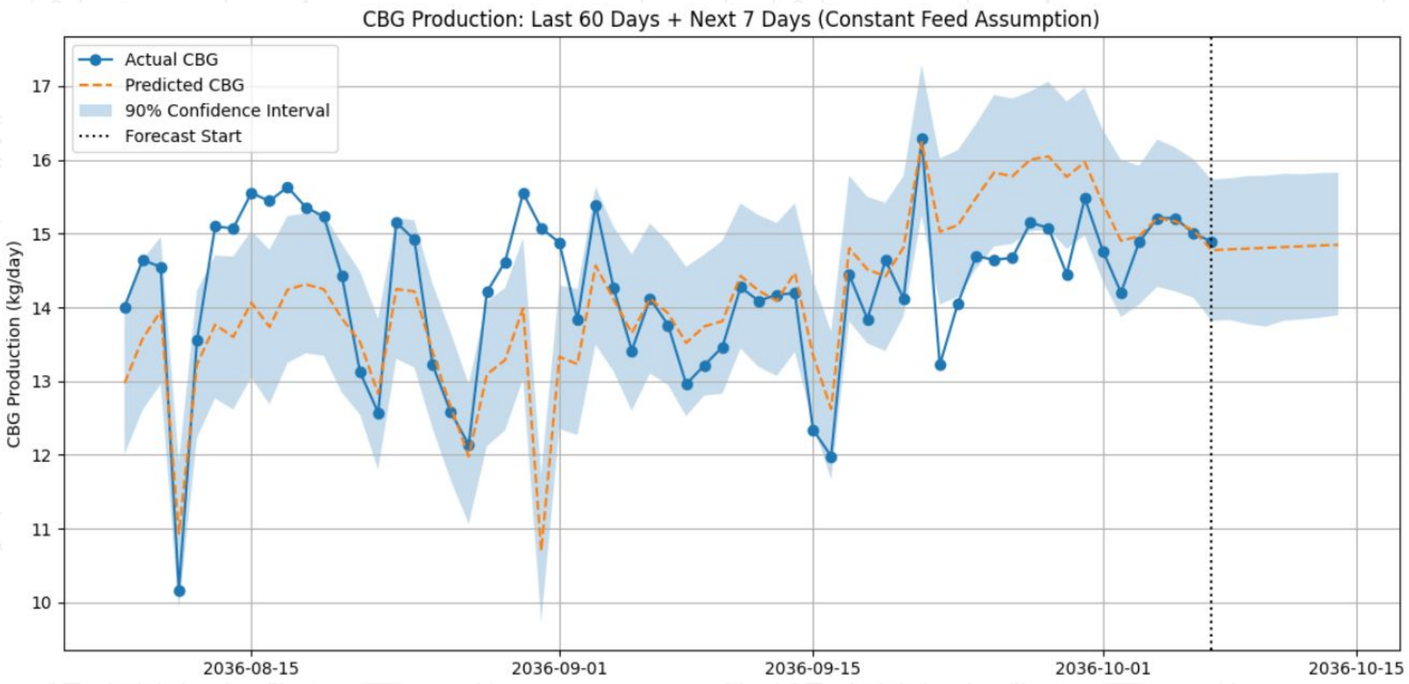
- Regressor 1
- Regressor 2
- Regressor 3

"Total Feed TPD",
"Total VS TPD",
"OLR",
"HRT",
"VFA",
"Raw biogas production (Nm3/day)",
"Slurry fed into digester (Nm3)",
"Slurry fed out of digester (Nm3)

Model Fitted

Training Data (df_model)

Result

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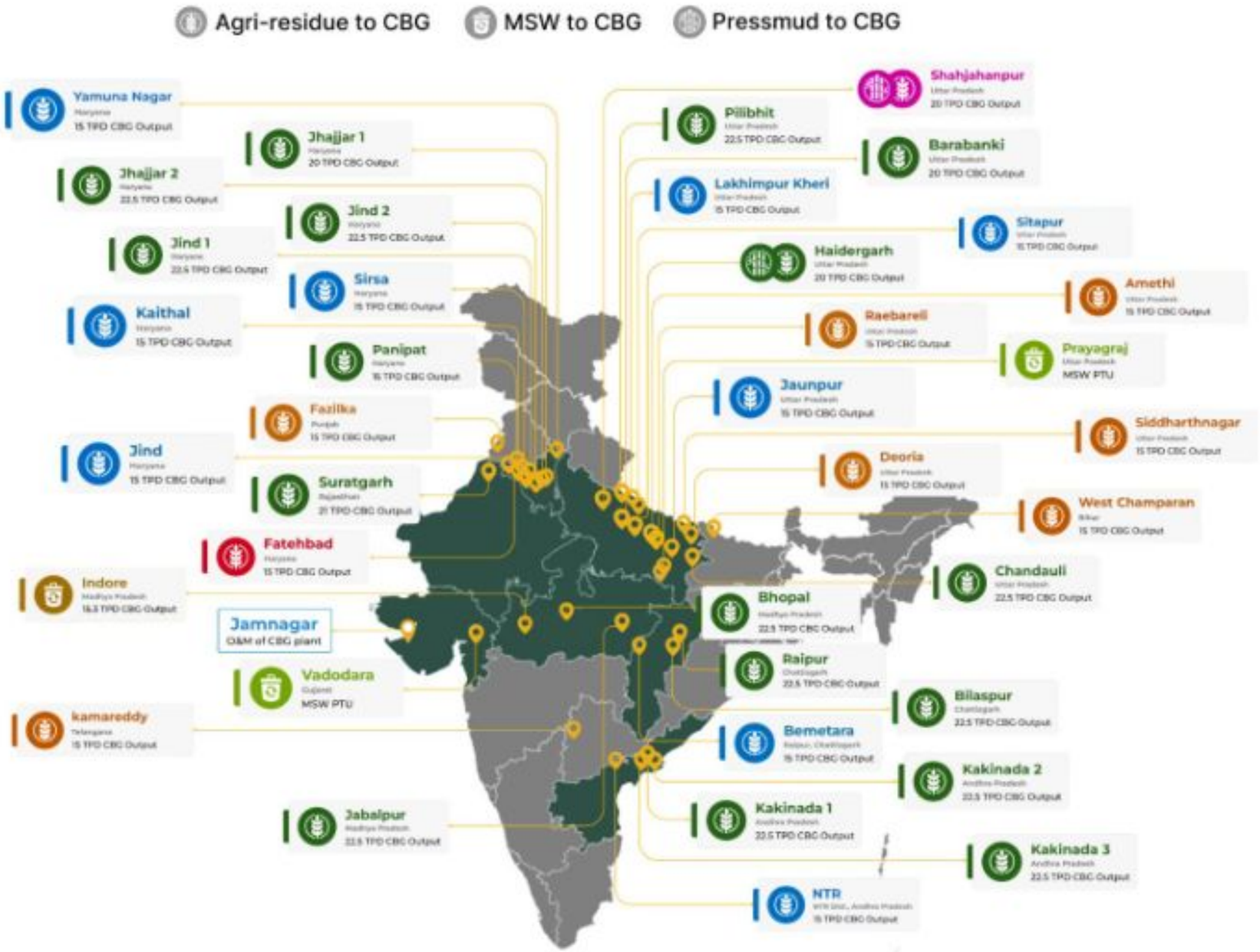
INFO:prophet:Making 79 forecasts with cutoffs between 2035-04-03 00:00:00 and 2036-09-30 00:00:00
100%|                                        79/79 [00:15<00:00,  1.75it/s]
  horizon       mse      rmse       mae      mape     mdape     smape  \
0  1 days  0.306018  0.553189  0.428074  0.076065  0.047905  0.079650
1  2 days  0.446364  0.668105  0.509075  0.074359  0.060279  0.073458
2  3 days  0.409623  0.640018  0.497753  0.080931  0.058295  0.079417
3  4 days  0.514680  0.717412  0.554149  0.085759  0.066743  0.085081
4  5 days  0.716426  0.846420  0.597139  0.096317  0.070961  0.097835

   coverage
0  0.848101
1  0.797468
2  0.784810
3  0.734177
4  0.721519
```

Predicted for 5 days with mse of 0.72 tons with a confidence of 72%.

How big we are ?

Our Projects

- Reliance Projects
- IGRPL Projects
- BGBPL Projects
- Ampyr Project
- MSW PTU
- Others

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